IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 9: Impacts of Transportation Systems-IV

Prof. Bhola Ram Gurjar

Dept. of Civil Engineering



Contents

- Impact of transportation on Soil and Water
- Impacts of Water Transport
- Impacts of Air Transport
- Acid Rain and its Impacts
- Mitigation measures
- Conclusion



Impact of transportation on Soil and Water

Physical Impacts

- The construction of transport infrastructure affects the physical environment of soil due to the need to clear, level, fill, and cut natural material.

- Construction work changes soil density, landscape relief, surface and groundwater flows, and microclimate, and thus alters land cover, vegetation and habitat composition.



Source: (Andreas Seiler, 2003)

Impact of transportation on Soil and Water

Physical Impacts



- Wetlands and habitats are especially sensitive to changes in hydrology
  - Ex. Changes caused by embankments and cuttings which may drain aquifers and increase the risk of soil erosion and extensive earth slides.
- Potential to pollute watercourses with sediments
- The canalisation of surface water into ditches can also significantly change water run-off and debris flows.

Source: (Andreas Seiler, 2003)

Impact of transportation on Soil and Water

Physical Impacts



Source: (World Bank, 2019)



Interrupted subsurface water flow due to road openings

Impact of transportation on Soil and Water

Chemical Impacts

- The Chemical impacts due to transportation infrastructure are caused by pollutants such as road dust, salt, heavy metals, fertiliser nutrients, and toxins.

- Pollutants generally accumulate in close proximity to the infrastructure but, in some cases, direct effects on vegetation and fauna can be observed at distances over several hundreds of metres away.

- Dust mobilised from the infrastructure, is transported and deposited along verges and in nearby vegetation.



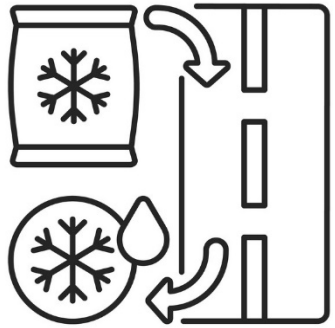
Source: (Andreas Seiler, 2003)

Impact of transportation on Soil and Water

Chemical Impacts

- De-icing and other salts (e.g. NaCl, $CaCl_2$, KCl, $MgCl_2$) can cause extensive damage to vegetation, contaminate drinking water supplies and reduce the pH-level in soil thus increasing the mobility of heavy metals.

- Heavy metals and trace metals (*e.g.* Pb, Zn, Cu, Cr, Cd, Al from petrol, de icing salts, and dust) can accumulate in plant and animal tissues and can affect their reproduction and survival rates.





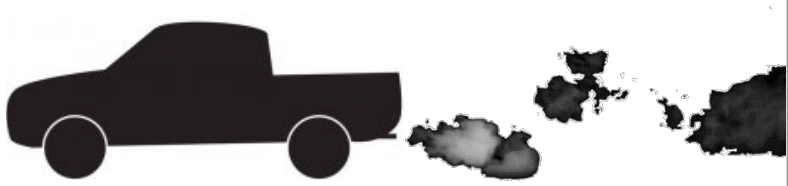
Source: (Andreas Seiler, 2003)

Impact of transportation on Soil and Water

Chemical Impacts

- Traffic exhaust emissions contain toxins such as polycyclic aromatic hydrocarbons (PAH), dioxins, ozone, nitrogen, carbon dioxide, and many fertilising chemicals.

- Traffic emissions in lakes and heathlands at a distance of over 200 metres away from the road can affect changes in plant growth and plant species diversity.



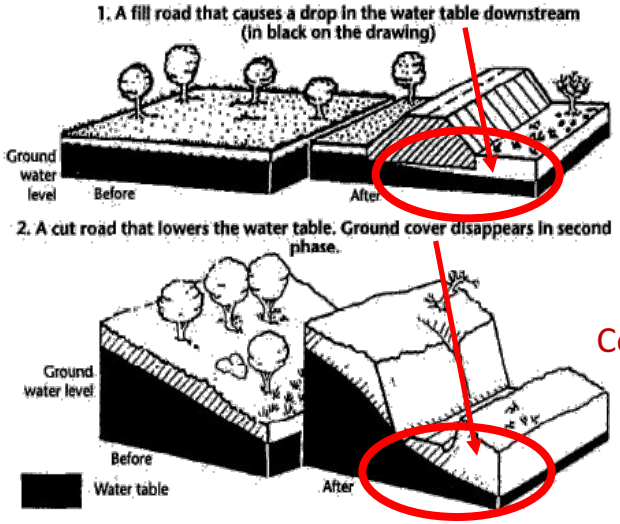
Source: (Andreas Seiler, 2003)

Impact of transportation on Water Quality

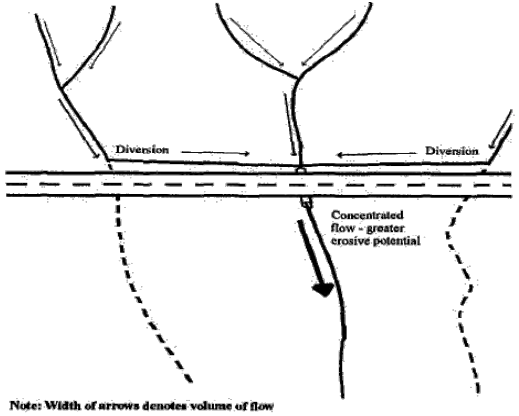
Groundwater flow modification

- Road drainage and excavation lowers the water table in surrounding areas, while embankments and structures raises the water table by restricting flow.
- Can contribute to flooding, soil erosion, channel modification, and siltation of streams.

Source: (FAO, 1998)



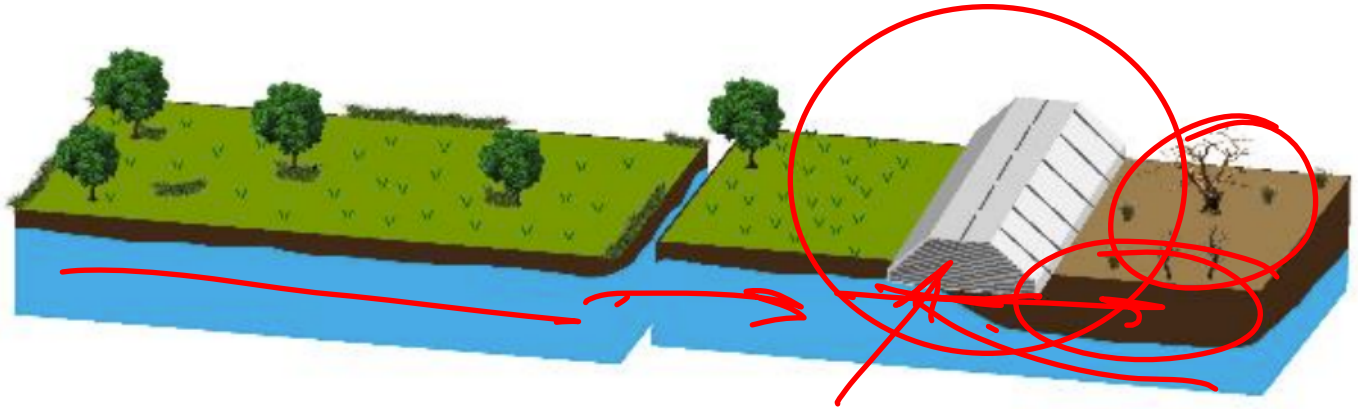
Modifications in Ground Water Table as a Result of Road Construction



Concentration of Surface Water Flow

Impact of transportation on Water Quality

Ground water flow Modification



Left: Ground water before Road construction

Right: Ground water after Road construction

- Road construction blocking subsurface streams, causing wetting upflow and drying downflow

Source: (World bank, 2019)

Impact of transportation on Water Quality

Inadequate Drainage on Roads



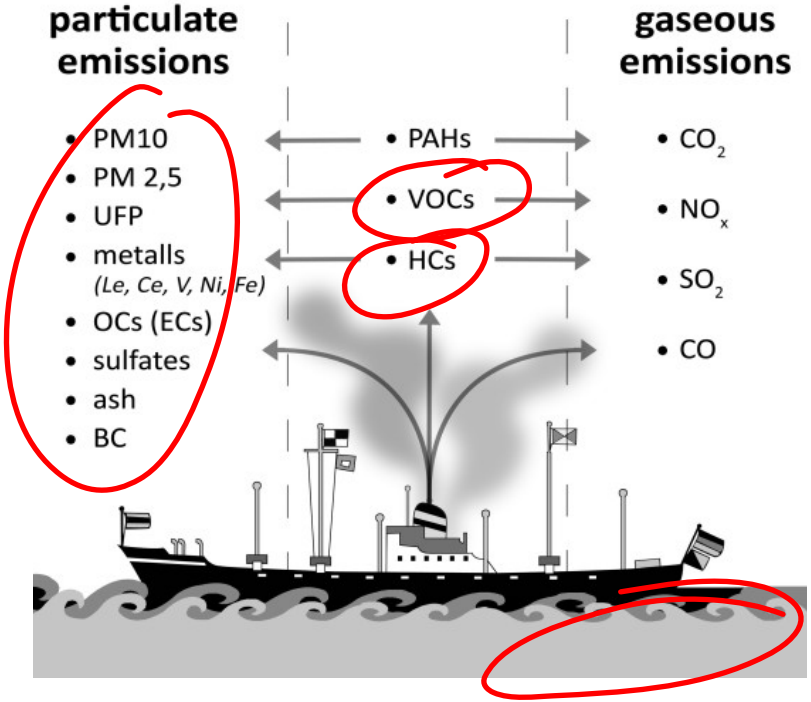
Improper and inadequate drainage systems restricts water flow, and can lead to rising water tables or flooding



Source: (World bank, 2019)

Impact of Water Transportation

Water Quality Degradation



Source: (FAO, 1998)

- Marine transport emissions represent the most important segment of water quality impact of the transportation sector.

Impact of Water Transportation

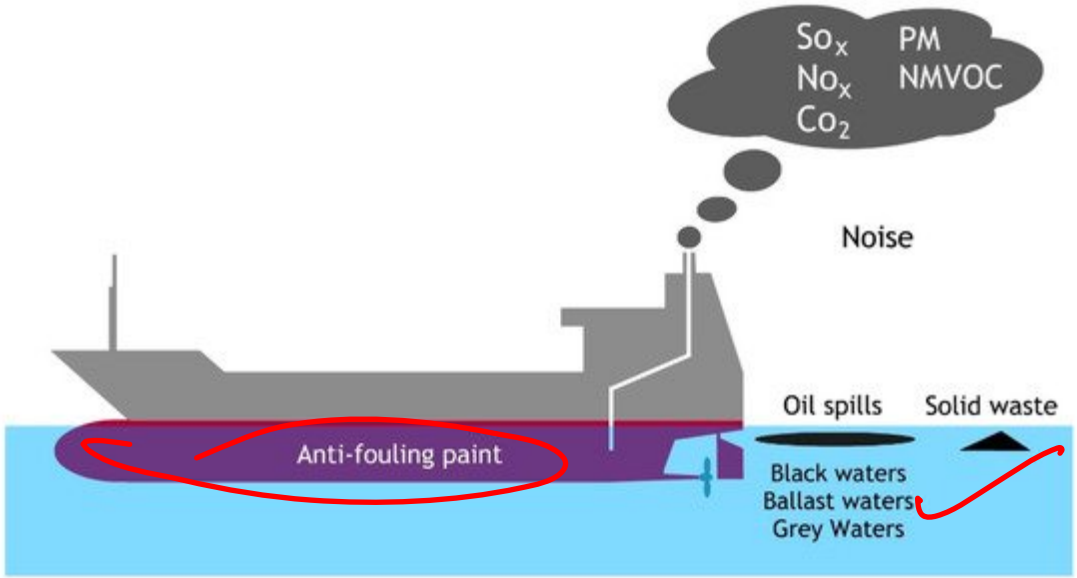
Water Quality Degradation

- The main effects of marine transport operations on water quality predominantly arise from dredging, waste, ballast waters and oil spills.
- Dredging is the process of deepening harbor channels by removing sediments from the bed of a body of water.
- Waste generated contain a very high level of bacteria that can be hazardous for public health as well as marine ecosystems when discharged in waters.



Source: (Jean Paul Rodrigue et. al., 2006)

Impacts Caused by Water Transport



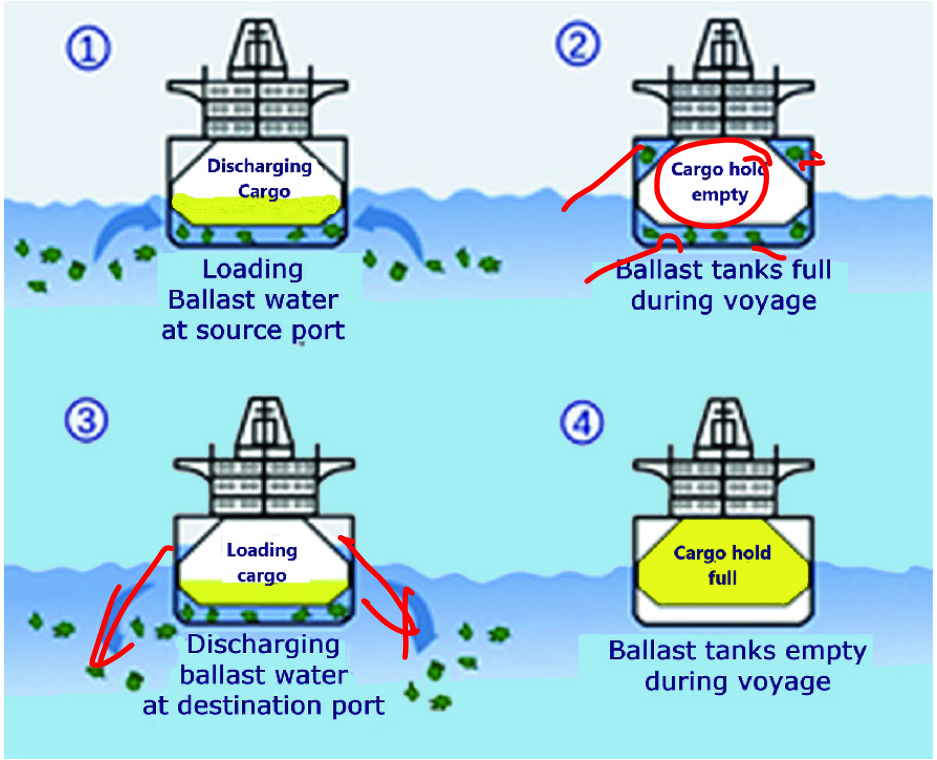
Typical Emissions from use of a marine vessel

Source: (Bruna Bacalja et. al., 2020)

Emissions from use of a marine vessel includes:

- Air pollutant emissions
- Water emissions such as
  - Cargo hold cleaning water
  - Ballast water
  - Grey water
  - Sewage
- Oil spills

Water Quality Impact from Water Transport



- Ballast water discharge from other sides of the world cause a certain type of pollution different from an emission type pollution
- It introduces invasive species that can cause domestic species to go extinct.

Source: (Komathy, K. 2019)

Water Quality Impact from Water Transport

Oil Spills



- Mostly associated with ship pollution.
- Less frequent than the pollution from daily operations, but oil spills have devastating effects.
- Toxic to marine life.

Source: (USEPA, 2012)

Water Quality Impact from Water Transport

Oil Spills



This photo taken by Georges de La Tremoille of Mu Press shows oil leaking from the MV Wakashio, a bulk carrier ship that ran aground off the southeast coast of Mauritius, August 7, 2020

- Polycyclic aromatic hydrocarbons (PAHs), the components in crude oil, are difficult to clean up, and last for years in the sediment and marine environment.
- Marine species constantly exposed to PAHs can exhibit developmental problems, susceptibility to disease, and abnormal reproductive cycles.

Source: (USEPA, 2012)

Water Quality Impact from Water Transport

Haven Oil Spill 1991, Italy



The very large crude carrier HAVEN, with 144,000 tonnes of Iranian heavy crude oil onboard exploded due to an electric spark during tank cleaning.

- Over 10,000 tonnes of oil have spilled into the ocean

Source: (ITOPF, https://www.itopf.org/in-action/case-studies/case-study/haven-italy-1991/)

Water Quality Impact from Water Transport

Gulsar Ana Oil Spill 2009, Madagascar



The 23,602 GT bulk carrier GULSAR ANA ran aground on the coast of Madagascar.

- Reported that 39,250 MT of cargo, 568 MT of heavy fuel oil, 66 MT of diesel oil and 8 MT of lube oils were lost to sea
- Shore clean up lasted 120 days with 7500 man-days of effort

Source: (ITOPF, https://www.itopf.org/in-action/case-studies/case-study/gulsar-ana-madagascar-2009/)

Water Quality Impact from Water Transport

RENA Oil Spill 2011, New Zealand



The 37,209 GT Containership RENA ran aground on Astrolabe Reef.

- The vessel had 1368 containers in which 11 were having dangerous goods and 1700 tonnes of IFO 380 bunker fuel and 61 tonnes diesel oil on-board.
- Estimated that approx. 200 tonnes of fuel oil were released.
- More than 1000 oiled birds and 250 live oiled birds were found including endangered penguin species.

Source: (ITOPF, https://www.itopf.org/in-action/case-studies/case-study/flinterstar-belgium-2015/)

Water Quality Impact from Air Transport

- Airports can generate significant water pollution due to their extensive use and handling of jet fuel, lubricants and other chemicals.

- In cold climates, the use of de-icing fluids can cause water pollution, as most of the fluids applied to aircraft subsequently fall to the ground and can be carried via storm water runoff to nearby streams, rivers or coastal waters, if not properly recovered.



(Source: US dept. of Federal Aviation Administration, FAA)

Source: (USEPA, 2012)

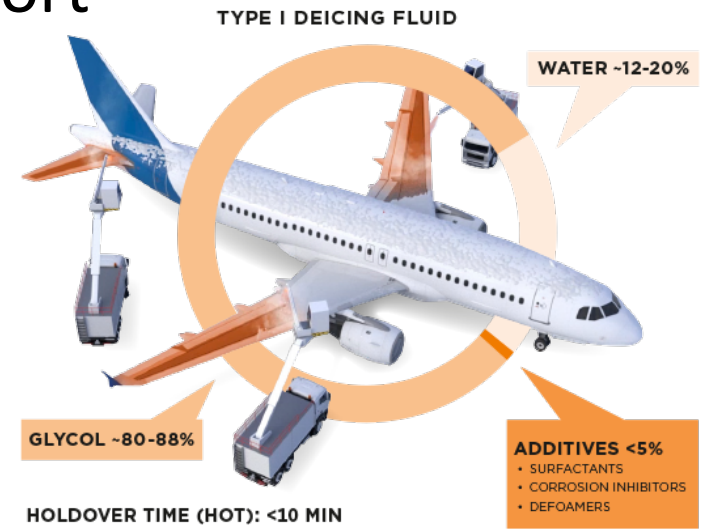
Water Quality Impact from Air Transport

- Airlines use deicing fluids based on ethylene glycol or propylene glycol as the active ingredient.

- Ethylene glycol and propylene glycol are known to exert high levels of biochemical oxygen demand (BOD) during degradation in surface waters.

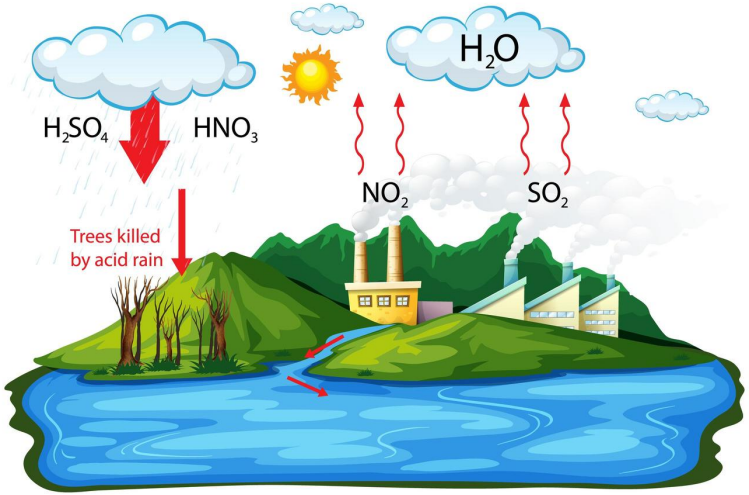
- Can adversely affect aquatic life by consuming oxygen needed by aquatic organisms for survival.

Source: (USEPA, 2012)



Acid Rain

Acid rain is formed when the acidic air pollutants such as Sulphur dioxide (SO_2) and Nitrogen dioxide (NO_2) dissolve in rain water.



- SO_2 dissolves in water to form Sulphurous acid (H_2SO_3).

Eq: $SO_2(g) + H_2O(l) \longrightarrow H_2SO_3(aq)$

- In presence of O_2 in the air, Sulphurous acid is slowly oxidized into Sulphuric acid (H_2SO_4).

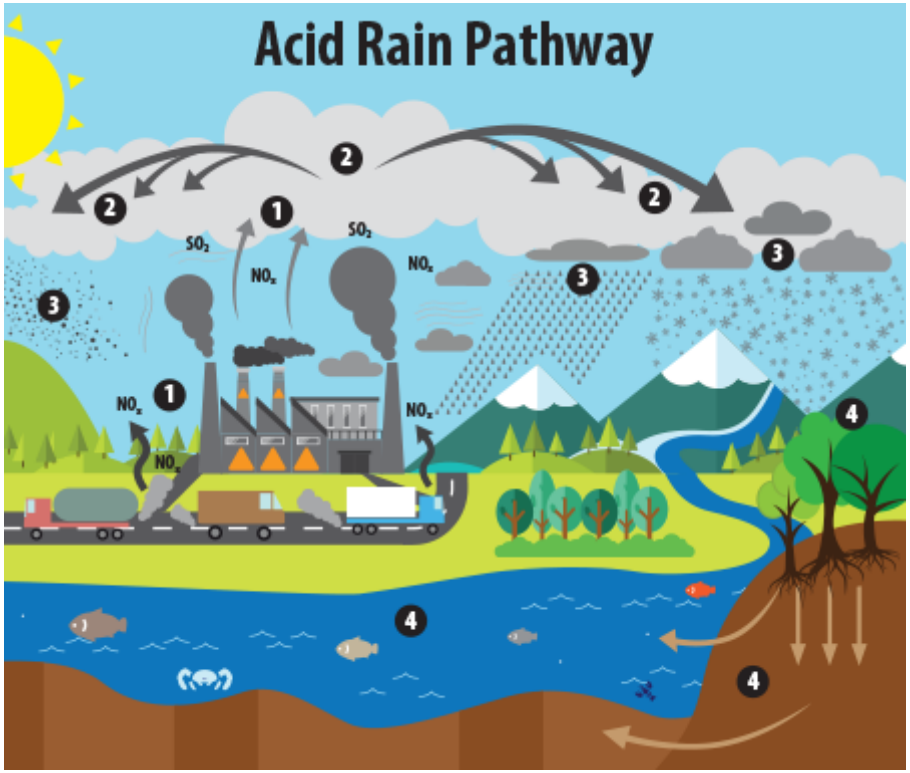
Eq: $H_2SO_3(aq) + O_2 \longrightarrow H_2SO_4$

- Similarly, In the presence of O_2 and H_2O, NO_2 is converted into Nitric acid (HNO_3).

Eq: $4NO_2(g) + 2H_2O \longrightarrow 4HNO_3$

Source: (USEPA, https://www.epa.gov/acidrain/what-acid-rain)

Acid Rain: Pathway



Source: (USEPA, https://www.epa.gov/acidrain/what-acid-rain)

Pathway for acid rain in our environment:

1. Emissions of SO_2 and NO_x are released into the air
2. The pollutants are transformed into acid particles that may be transported long distances
3. These acid particles then fall to the earth as wet and dry deposition (dust, rain, snow, etc.)
4. May cause harmful effects on soil, forests, streams, and lakes.

Acid Rain: Effects on Land/ Soil



- Effects soil and vegetation on land/ soil.
- Causes soil's acid levels to rise and this kills plants growing on it.

Source: (https://www.conserve-energy-future.com/causes-and-effects-of-acid-rain.php)

Acid Rain: Effects on Plants

- Stops trees and plant life from receiving Oxygen thus stopping them from growing.







Source: (U.S Geological survey, USGS)

Acid Rain: Effects on Material Assets



- Affects Buildings and Sculptures
- Dissolves lime stone and marble and could erode or even destroy sculptures.

- Causes lime stone and marble to erode causing buildings to wither away.



Source: (https://acidraingermany.weebly.com/acid-rain-in-germany.html)

Acid Rain: Effects on Water Bodies

- Lakes can be acidified easily due to low water flow.
- Most of aquatic species cannot survive in acidified lakes and hence affect biodiversity.

Critical pH Levels for Aquatic Organisms

Animal	Critical pH Level
Snails	6
Clams	6
Bass	5.5
Crayfish	5.5
Mayfly	5.5
Trout	5
Salamanders	5
Perch	4.5
Frogs	4



Source: (USEPA, https://www.epa.gov/acidrain/what-acid-rain)

Acid Rain: Effects on Humans

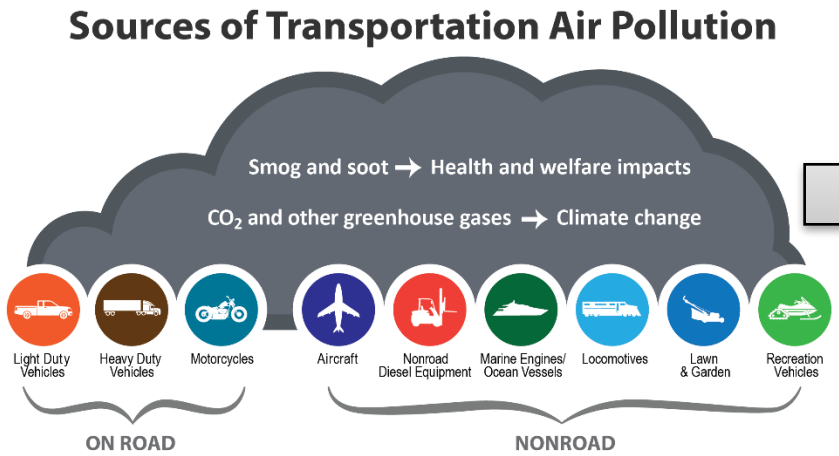
Only Indirect effects on Humans

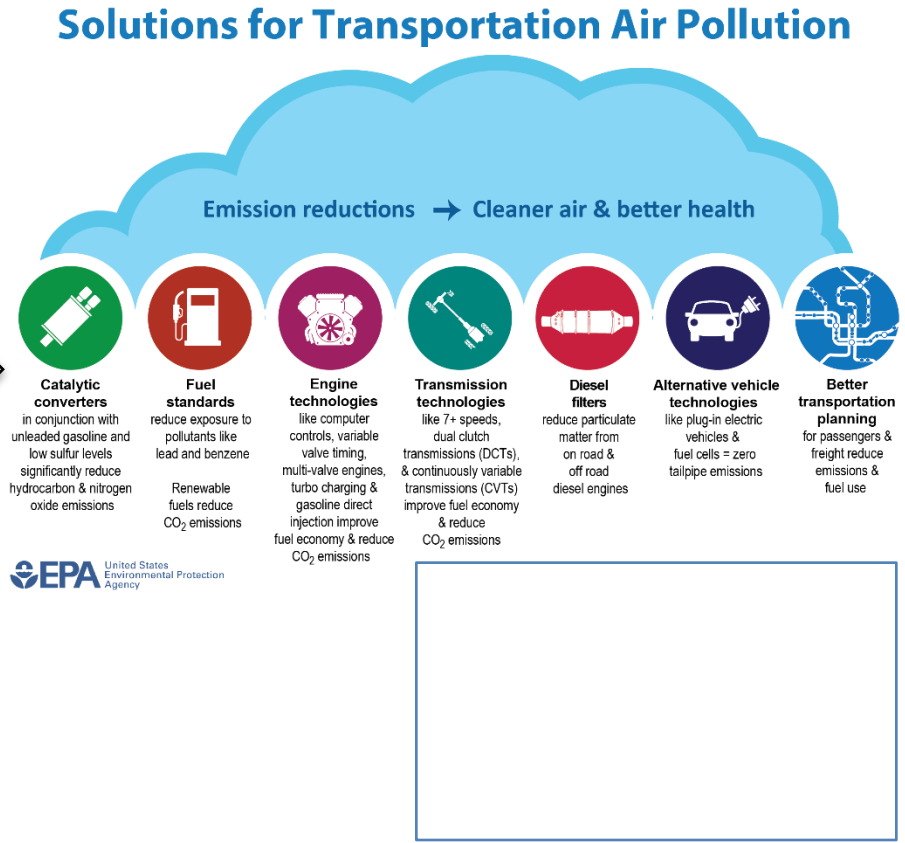
- Sulphur dioxide (SO_2) particles in the air can encourage chronic lung problems, like asthma and bronchitis.
- Nitrogen oxides (NO_X) can promote formation of Ground-level Ozone, which promotes severe lung problems like chronic pneumonia and emphysema.



Source: (Robert Boumis, 2019)

Sources and Solutions for Transport Air Pollution

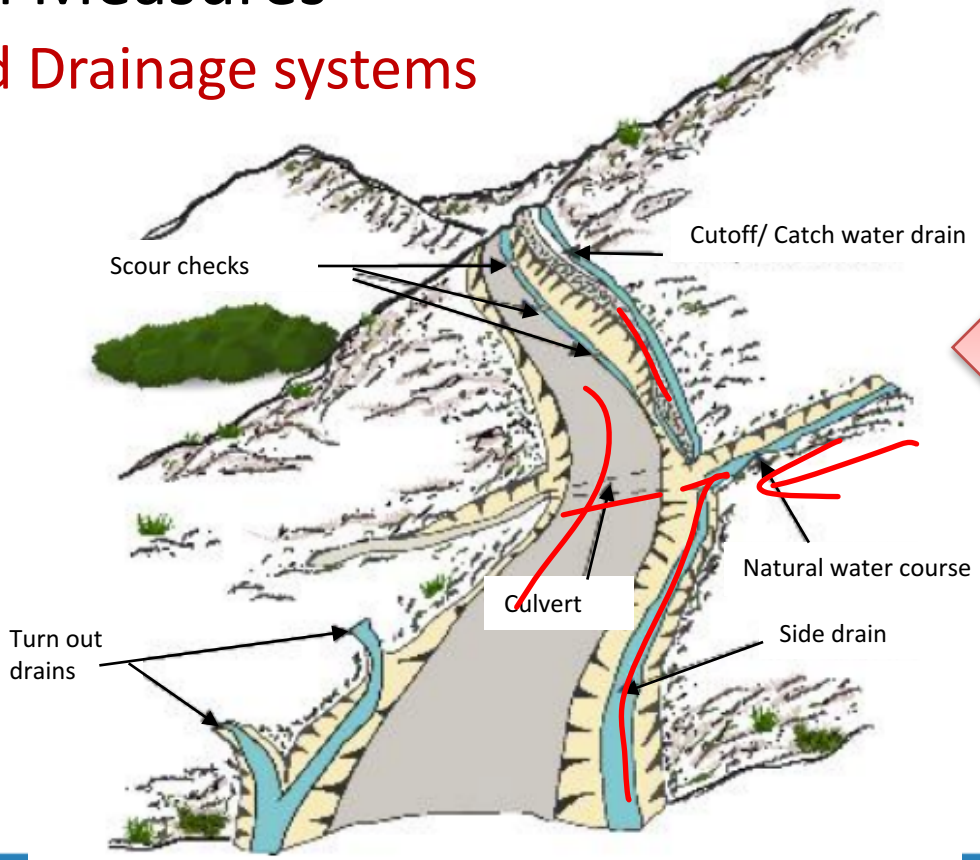




Source: (USEPA, https://www.epa.gov/transportation-air-pollution-and-climate-change/learn-about-air-pollution transportation)

Mitigation Measures

Good Road Drainage systems



Source: (World Bank, 2019)

Features of a Good Road drainage system

Mitigation Measures

Storm water Inlets and Sewers





Storm water inlets and sewers to remove water from road surfaces.

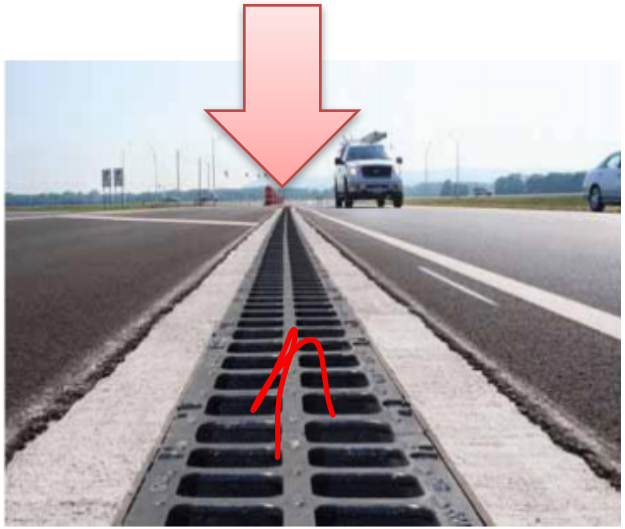
Source: (Dipanjan Mukherjee, 2014)

Mitigation Measures

Highway Surface drainage systems



Storm water drainage systems on sides and middle of roads



Source: (Dipanjan Mukherjee, 2014)

Mitigation Measures

Infiltration Structures from Road Drainage





- Cross and Side drains to evacuate water away from Road structures

Water guided to Ground water recharge structures

Source: (World Bank, 2019)

Mitigation Measures

Roadside Infiltration Trenches



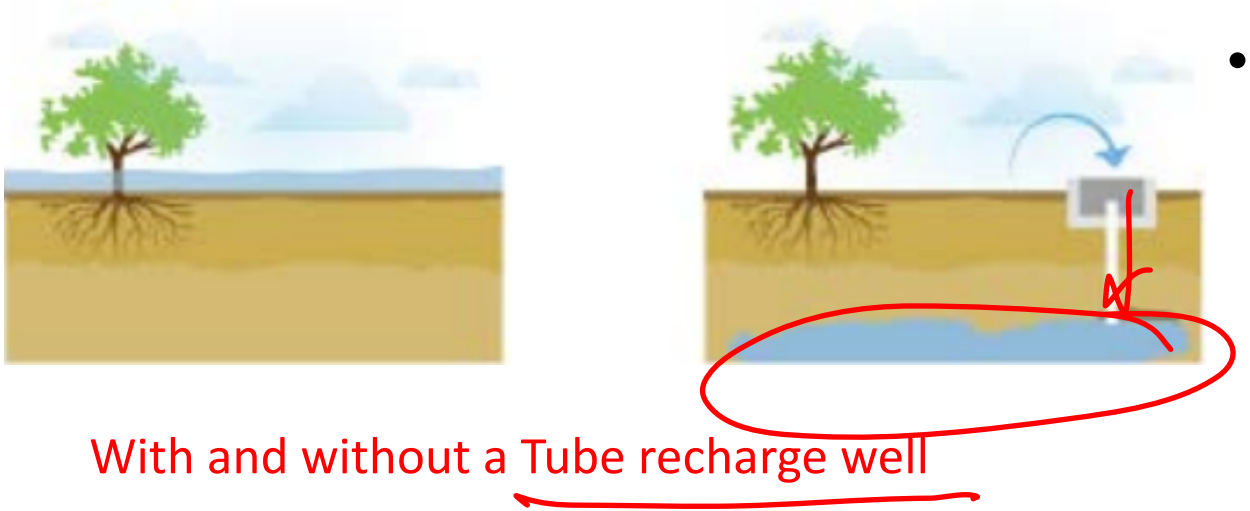
Source: (World Bank, 2019)

- Series of Roadside infiltration trenches with bunds to intercept additional surface runoff

Roadside infiltration trenches with bunds

Mitigation Measures

Tube Recharge wells or Bhungroo



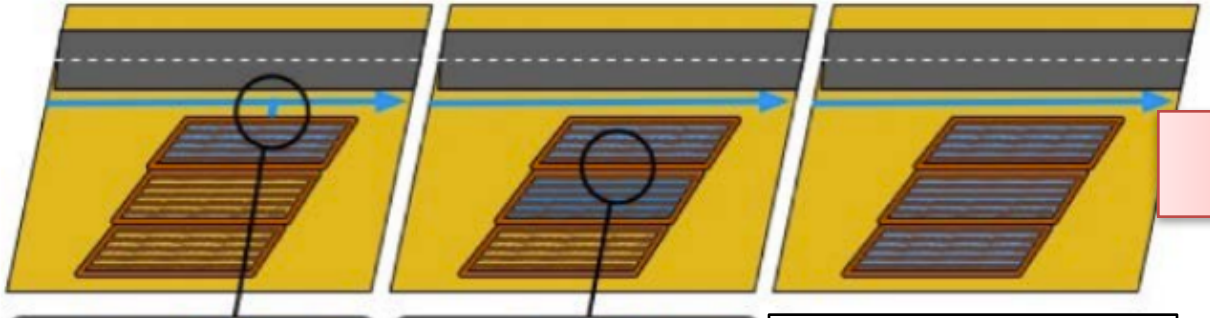
With and without a Tube recharge well

Source: (World Bank, 2019)

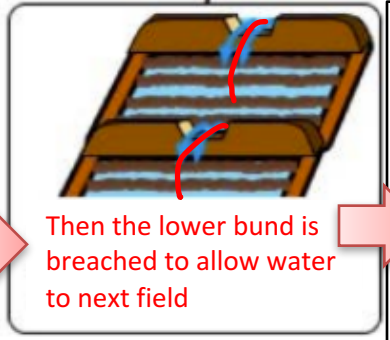
- Tube recharge well collecting excess water for recharge
- Known as Bhungroo in Gujarat, India

Mitigation Measures

Irrigation from Road drainage



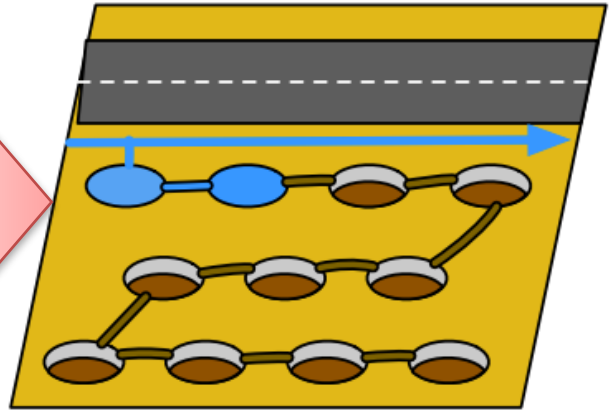
The upper bund is breached to let water flow in the first field



Then the lower bund is breached to allow water to next field

- The operation is repeated until all fields are watered
- Important to have smooth water movements to avoid erosion
- Fields should be well levelled

- Runoff is directed to interconnected soaking pits



Source: (World Bank, 2019)

Mitigation Measures

Road water harvesting



Road water harvesting from surface storage

Source: (World Bank, 2019)

Mitigation Measures

Irrigation from Road drainage



Roadside borrow pit for Rice drainage in Bangladesh

Source: (World Bank, 2019)

Mitigation Measures

Irrigation from Road drainage



Growing grapes with road water in Bolivia

Source: (World Bank, 2019)

Mitigation Measures

Irrigation from Road drainage



Collecting road water for drainage in Shandong, China

Source: (World Bank, 2019)

Mitigation Measures

Urban lakes



Source: (World Bank, 2019)

"Ijzeren Vrouw" is an Urban lake in Netherlands, converted from Borrow pit

Oil Spill Clean-up

Oil Booms or Containment Booms



- Oil booms are temporary floating barriers
- Reduces the risk of spreading of oil spills to shorelines, rivers and oceans

Oil Spill Clean-up

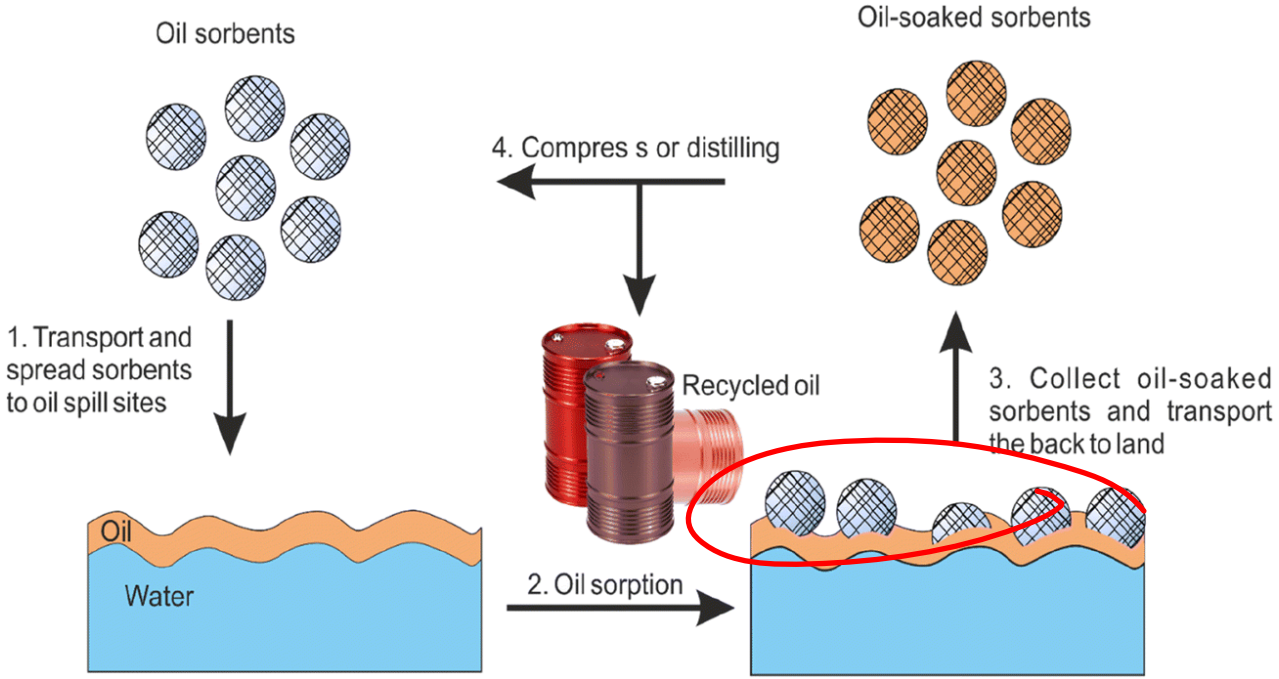
Oil Skimmers



- Oil skimmers recovers spilled oil from surface of water.
- Traps the oil floating on top of water surface.

Oil Spill Clean-up

Oil Sorbents

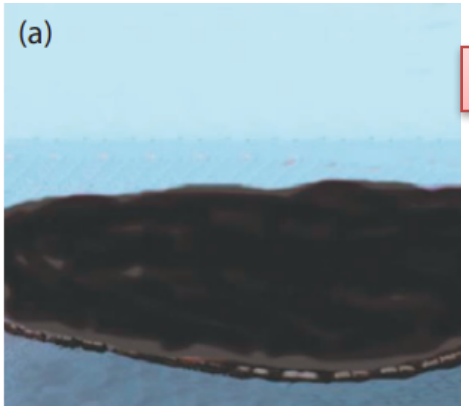


Source: (https://link.springer.com/article/10.1007/s11356-021-13775-z/figures/9)

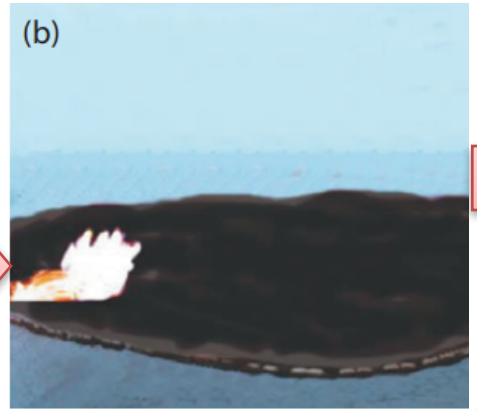
- Sorbents absorb oil from the water surface
- Includes clay or vermiculates or other synthetic sorbents such as plastic or fibres.

Oil Spill Clean-up

Burning In-Situ



(a) Oil collected and held by a fire-resistant boom within a segregated area.



(b) Ignition-flame applied



(c) Flame spreads over and burns oil down to 1mm thickness

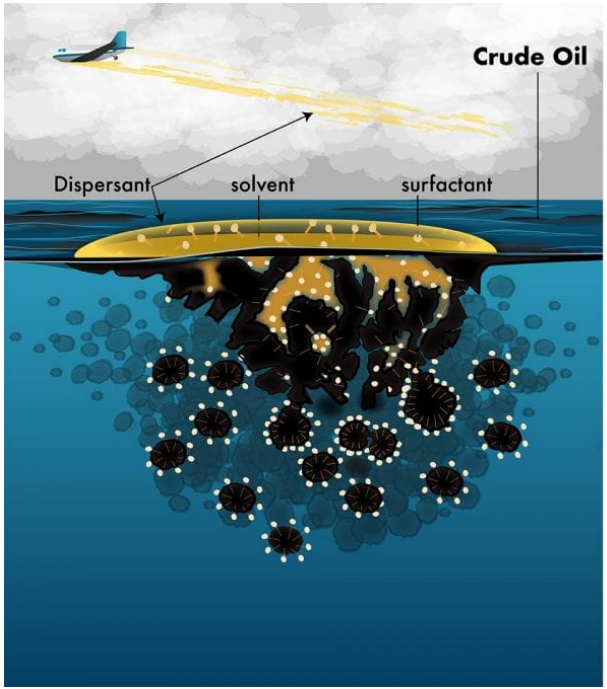


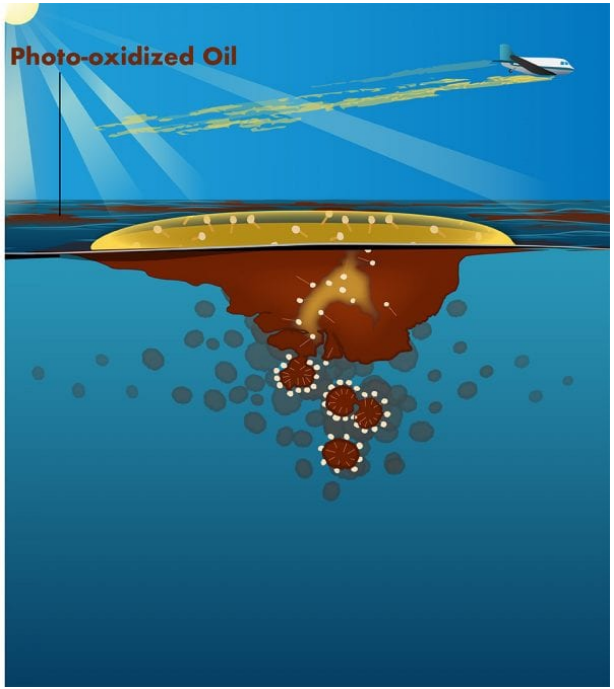
(d) When vapour is insufficient to maintain combustion, the fire ends

Source: (https://www.giwacaf.net/en/publications/in-situ-burning-ipieca-gpg)

Oil Spill Clean-up

Using Dispersants





- Dispersants contains a form of detergents
- Helps to break oil into smaller droplets and can become diluted in the ocean.

Source: (https://www.whoi.edu/oceanus/feature/reassessing-guidelines-for-oil-spill-cleanups/)

Oil Spill Clean-up

Manual labour



- Time taking process
- Can take months of manual work

Oil Spill Clean-up

Hot water and High pressure washing



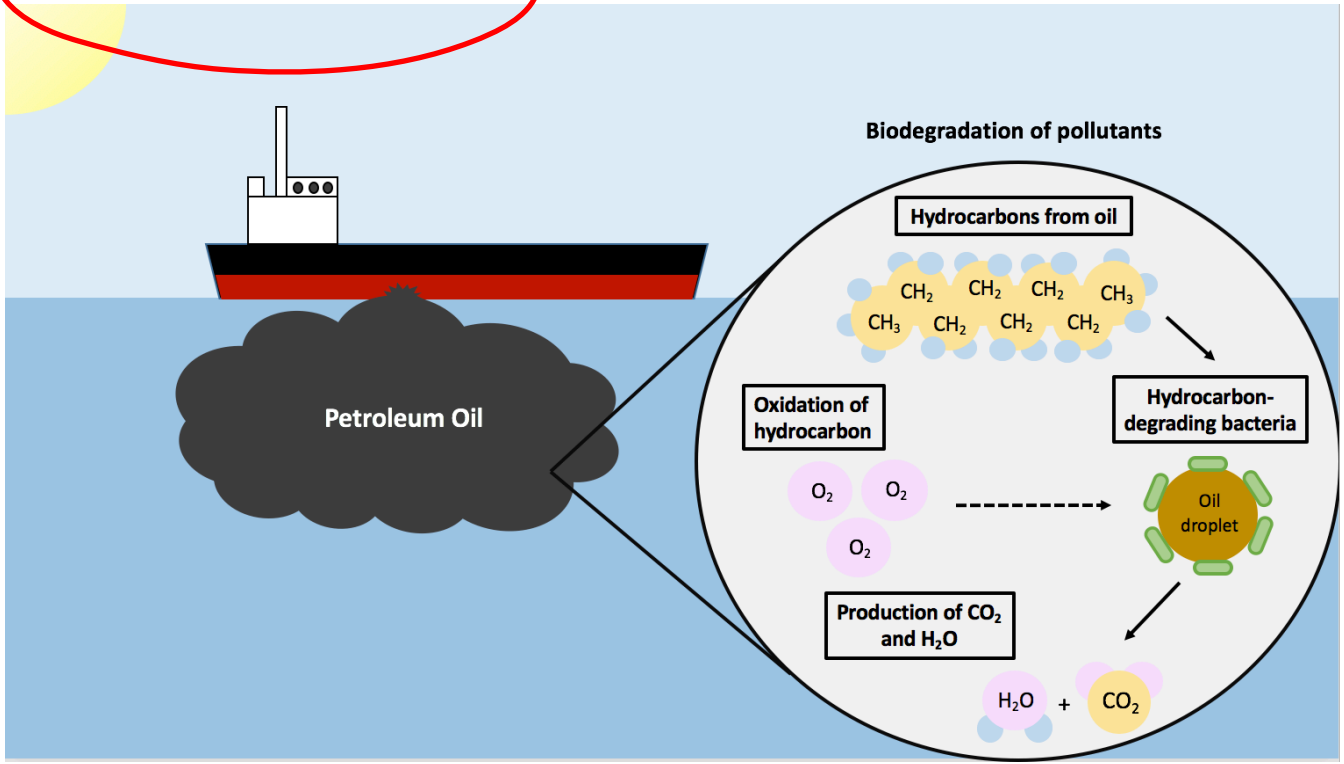
- Applied in shorelines
- Removes stranded oil from hard surfaces like rocks and sea walls
- Effective, but can harm plants and animals in the treated zone

Source: (https://response.restoration.noaa.gov/oil-and-chemical-spills/significant-incidents/exxon-valdez-oil-spill/high-pressure-hot-water-washing)

Oil Spill Clean-up

Bioremediation

Microbial bioremediation of oil spills



- Microorganisms use HC as carbon source for metabolism.
- HC are oxidized to form CO_2 and H_2O

Conclusion

- Transportation activities can cause physical and chemical impacts on Soil, water and other materials.
- The Marine and the Aviation industry can also significantly damage the environment, similar to the Road transportation systems.
- Sustainable measures should be adopted in the Transport sector to minimize the environmental impacts.
- Measures such as the use of low sulphur fuels and other alternate fuels significantly reduces the SO_X and NO_X emissions, which causes the Acid rain.



References

- Andreas Seiler, (2003). "*COST Action 341 - Habitat Fragmentation due to transportation infrastructure: The European Review*", Chapter-3, Effects of Infrastructure on Nature, Office for Official Publications of the European Communities, Luxembourg. pp. 31-50.
- Bruna Bacalja, Maja Krcum and Merica Sliskovic, (2020). "*A Line Ship Emissions while Manoeuvring and Hotelling—A Case Study of Port Split",* Journal of Marine Science and Engineering, Vol. 8, Issue 11, DOI. 10.3390/jmse8110953.
- Dipanjan Mukherjee, (2014). "*Highway Surface Drainage system & Problems of Water Logging in Road section*", The International Journal Of Engineering And Science (IJES), Vol. 3, Issue 11, pp. 44-51, ISSN (e): 2319 – 1813 ISSN (p): 2319 – 1805.
- Food and Agricultural Organization (FAO), (1998). "*Watershed management field manual",* FAO Conservation Guide 13/5, Food and Agricultural Organization of the United Nations, Rome, http://www.fao.org/3/T0099E/T0099E00.htm.
- Jean Paul Rodrigue, Claude Comtois and Btian Slack (2006). "*The Geography of Transport Systems*" Third Edition, New York: Routledge, Taylor and Francis Group, ISBN: 978–0–415–82253–4 (hbk), ISBN: 978–0–415–82254–1 (pbk), ISBN: 978–0–203–37118–3 (ebk), https://transportgeography. org/ wp-content/uploads/GTS_Third_Edition.pdf.
- Robert Boumis, (2019). "*Negative Health Effects of Acid Rain on Humans*", https://sciencing.com/negative-health-effects-acid-rain-humans-24007.html.
- World bank, (2019). "*Guideline: Green Roads for Water*", Road Infrastructure in Support of Water Management and Climate Resilience, https://roadsforwater.org/wp-content/uploads/2019/11/Guidelines-roads-for-water-Draft-vs8_2.pdf.
- United States Environmental Protection Agency (USEPA), 2012). "*Airport Deicing Effluent Guidelines*", https://www.epa.gov/eg/airport-deicing-effluent-guidelines-documents.

Thank You